PHYS 230 - Electricity and Magnetism
Lecture 13

Instructor: Saeed Rastgoo

UNIVERSITY
OF ALBERTA

Outline

3. Electric Potential Energy and Potential
   - Electric Potential

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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- V tells us if we were to put a charge q around the source Q, what would have been the potential energy of the system U_{Qq}
- Similar to **E** and **F**: Field **E** is property of just the source Q; tells us if a charge q is put in **E** of source, what would the force **F** be

Potential and Electric Potential

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- The potential difference $V_b - V_a$ is the negative of the work done by $\mathbf{E}$ of the source on a test charge, when the test charge moves from a to b, per unit charge

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Potential of A Point Charge (Using U)

Using the general formula for V and the point charge formula for U

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where $r_i =$ the distance from q_i to the point where we are computing V_i

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- This is a unit of energy! Not potential difference

Important Point!

- In any computation which involves an integral over a curve with the infinitesimal displacement $d\mathbf{r}$, regardless of the direction of the curve, DO NOT CHANGE THE DIRECTION (SIGN) OF $d\mathbf{r}$.
- **Change of integral limits** due to change of direction of movement on the curve **will take care of the sign!**

Exercise 1

https://epoll.srv.ualberta.ca – Code: MBD

A proton (charge $+\mathrm{e} = 1.602 \times 10^{-19}$ C) moves a distance $L = 0.50$ m in a straight line between points a and b in a linear accelerator. The electric field is uniform along this line, with magnitude $E = 1.5 \times 10^{7}$ V/m in the direction from a to b. Determine the potential difference $V_b - V_a$.

Exercise 1

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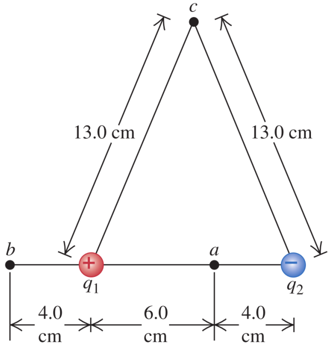
Exercise 2 https://epoll.srv.ualberta.ca – Code: MBD

An electric dipole consists of point charges $q_1 = +12$ nC and $q_2 = -12$ nC placed 10.0 cm apart as in figure. Compute the electric potentials at points *a*, *b*, and *c*.



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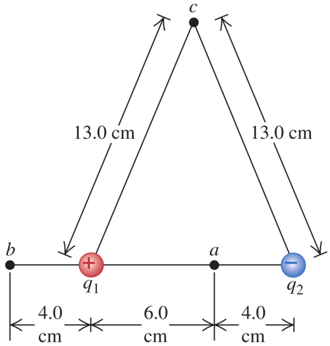
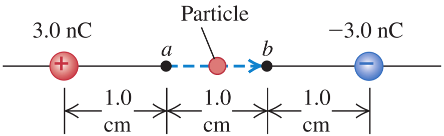


Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD

In this figure, the left charge is $q_1 = 3$ nC and the right one is $q_2 = -3$ nC. A dust particle (test charge) with mass $m = 5.0 \times 10^{-9}$ kg and charge $q_0 = 2.0$ nC starts from rest and moves in a straight line from point a to point b. What is its speed v at point b? **For a test charge, we ignore its effects, e.g., computing potentials V, we just consider the potentials due to q_1 and q_2.**



Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD